

Database Schema & Entity Models

Technical specification for the Document Review & AI Compliance Platform with zero-knowledge PII vault and RAG vector corpus.

ORM: SQLAlchemy 2.0 (Declarative Base)

Engine: SQLite 3 / PostgreSQL 14+

Key Standard: UUIDv4 (36-char)

Total Entities: 10 Tables (6 Enums)

1. Architectural Domains

• Authentication & Users

Role-based access control separating Advisors (authors) and Compliance Officers (reviewers).

users

notifications

• Document & Revisions

Hierarchical document storage with self-referential revision threads (`thread_id` lineage).

documents

review_decisions

• AI Compliance Engine

Automated pre-review violation scans linking extracted passages with SEC/FINRA rule criteria.

ai_analyses

compliance_flags

• Audit, PII Vault & RAG Corpus

Immutable audit trail, isolated local PII de-identification vault, and vectorized regulatory rules.

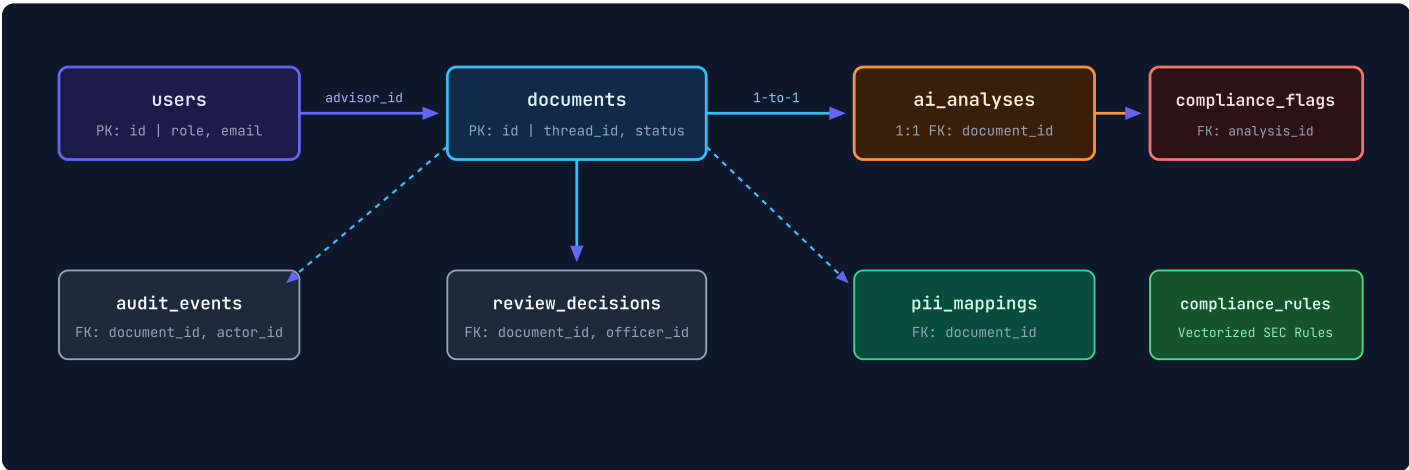
audit_events

pii_mappings

compliance_rules

precedent_submissions

Entity Relationship Overview (ERD)



2. Enumerations & Core Tables

Statically Enforced Enums

Enum Type	Allowed Values	Description / Usage
UserRole	advisor , officer	Access tier gating submission vs officer review portals
DocumentStatus	pending_review , approved , rejected , needs_revision	Document workflow state & precedent decisions
DocumentType	marketing_email , brochure , social_post , meeting_notes , proposal_letter , other	Filing classification for contextual prompt selection
AuditAction	submitted , viewed , decided , resubmitted , analysis_generated , downloaded	Immutable compliance trail action taxonomy
FlagSeverity	critical , high , medium , low	AI-detected compliance risk score
RuleCategory	required_disclosure , prohibited_claims , performance_standards , supervision	SEC/FINRA regulatory domain indexing

1. users

User accounts, roles, and authentication credentials

Column	Type	Modifiers	Description
id	VARCHAR(36)	PK NN	UUIDv4 primary key
email	VARCHAR(255)	UQ IDX NN	User unique login email address
hashed_password	VARCHAR(255)	NN	Bcrypt hashed password string
full_name	VARCHAR(255)	NN	User display / legal name
role	Enum(UserRole)	NN	advisor or officer
created_at	DATETIME	NN	Account creation timestamp (UTC)

2. documents

Compliance submissions, metadata, extracted text, and revision links

Column	Type	Modifiers	Description
id	VARCHAR(36)	PK NN	UUIDv4 document primary key
advisor_id	VARCHAR(36)	FK USERS.ID IDX NN	Authoring advisor account ID
title	VARCHAR(255)	NN	Document submission title
document_type	Enum(DocumentType)	NN (Default: other)	Classification type (marketing email, brochure, etc.)
status	Enum(DocumentStatus)	IDX NN (Default: pending_review)	Workflow status (pending, approved, rejected, revision)
file_name	VARCHAR(255)	NN	Original uploaded filename
file_path	VARCHAR(512)	NN	Absolute/relative path on disk
file_size	INTEGER	NN	File size in bytes
mime_type	VARCHAR(100)	NN	MIME format (application/pdf, text/plain, etc.)
extracted_text	TEXT	OPT	Plaintext parsed from document file
thread_id	VARCHAR(36)	IDX NN	Groups all revisions of the same submission
parent_document_id	VARCHAR(36)	FK DOCUMENTS.ID IDX OPT	Preceding document ID in version sequence
version_number	INTEGER	NN (Default: 1)	Version sequence counter
uploaded_at	DATETIME	NN	UTC submission timestamp
updated_at	DATETIME	NN	UTC last modified timestamp

3. Review & AI Compliance Analysis

3. review_decisions			Compliance officer final verdicts, notes, and timestamps
Column	Type	Modifiers	Description
id	VARCHAR(36)	PK NN	UUIDv4 decision primary key
document_id	VARCHAR(36)	FK DOCUMENTS.ID IDX NN	Reviewed document ID
officer_id	VARCHAR(36)	FK USERS.ID IDX NN	Compliance Officer who performed review
status	Enum(DocumentStatus)	NN	Decision state: approved , rejected , needs_revision
comment	TEXT	NN	Officer feedback, reasoning, and instructions
decided_at	DATETIME	NN	Timestamp of decision

4. ai_analyses			1-to-1 AI evaluation summaries, model metadata, and sanitized payload audit
Column	Type	Modifiers	Description
id	VARCHAR(36)	PK NN	UUIDv4 AI analysis primary key
document_id	VARCHAR(36)	FK DOCUMENTS.ID UQ IDX NN	1-to-1 foreign key to analyzed document
summary	TEXT	NN	Executive summary generated by LLM
outbound_payload_masked	TEXT	OPT	Audit snapshot of de-identified payload sent to AI
status	VARCHAR(50)	NN (Default: completed)	Status: completed , degraded , or failed
model_used	VARCHAR(100)	NN (Default: gemini-1.5-flash)	Model version used for analysis
generated_at	DATETIME	NN	Completion timestamp

5. compliance_flags			Granular risk violations, rule citations, excerpts, and suggested fixes
Column	Type	Modifiers	Description
id	VARCHAR(36)	PK NN	UUIDv4 violation flag primary key
analysis_id	VARCHAR(36)	FK AI_ANALYSES.ID IDX NN	Parent AI analysis record ID
passage_excerpt	TEXT	NN	Exact text from document triggering violation
matched_rule_id	VARCHAR(100)	OPT	Regulatory rule code (e.g. SEC-MKT-01)
matched_rule_title	VARCHAR(255)	OPT	Rule title / header
matched_rule_text	TEXT	OPT	Full statutory requirement text
explanation	TEXT	NN	Detailed explanation why passage is non-compliant
severity	Enum(FlagSeverity)	NN (Default: medium)	critical , high , medium , low
suggested_fix	TEXT	OPT	Remediation recommendation / replacement phrasing

4. Audit Trail, Notifications & PII Vault

6. audit_events				Immutable regulatory ledger tracking all interactions and state changes
Column	Type	Modifiers	Description	
id	VARCHAR(36)	PK NN	UUIDv4 audit event primary key	
document_id	VARCHAR(36)	FK DOCUMENTS.ID IDX NN	Document subject of the action	
actor_id	VARCHAR(36)	FK USERS.ID IDX OPT	User ID who triggered event (NULL for system)	
actor_name	VARCHAR(255)	OPT	Actor display name snapshot at time of event	
actor_role	Enum(UserRole)	OPT	Actor role snapshot	
action	Enum(AuditAction)	NN	Action type: <code>submitted</code> , <code>viewed</code> , <code>decided</code> , etc.	
details_json	TEXT	OPT	JSON encoded event metadata / payload diff	
created_at	DATETIME	IDX NN	Immutable timestamp (UTC)	

7. notifications				In-app notifications for review alerts, revisions, and status updates
Column	Type	Modifiers	Description	
id	VARCHAR(36)	PK NN	UUIDv4 notification ID	
user_id	VARCHAR(36)	FK USERS.ID IDX NN	Recipient user account ID	
document_id	VARCHAR(36)	FK DOCUMENTS.ID IDX OPT	Linked document ID (if applicable)	
title	VARCHAR(255)	NN	Notification header / title	
message	TEXT	NN	Detailed notification message text	
is_read	BOOLEAN	NN (Default: <code>false</code>)	Unread / read acknowledgment status	
created_at	DATETIME	IDX NN	Notification trigger timestamp	

8. pii_mappings				PII de-identification vault mapping tokens to real sensitive strings
Column	Type	Modifiers	Description	
id	VARCHAR(36)	PK NN	UUIDv4 mapping primary key	
document_id	VARCHAR(36)	FK DOCUMENTS.ID IDX NN	Document containing the redacted PII	
placeholder	VARCHAR(100)	NN	De-identified placeholder token (e.g. <code>[CLIENT_1]</code>)	
original_value	TEXT	NN	Original sensitive raw string (e.g. <code>John Doe</code>)	
entity_type	VARCHAR(50)	NN	Entity type: <code>name</code> , <code>email</code> , <code>ssn</code> , <code>amount</code>	
created_at	DATETIME	NN	De-identification timestamp	

5. Vector & RAG Corpus & Cascades

9. compliance_rules		SEC/FINRA regulatory rules and disclosure standards with vector embeddings	
Column	Type	Modifiers	Description
id	VARCHAR(36)	PK NN	UUIDv4 rule primary key
rule_code	VARCHAR(50)	UQ IDX NN	Unique statutory code (e.g. SEC-MKT-01)
category	Enum(RuleCategory)	IDX NN	Taxonomy: required_disclosure , prohibited_claims
title	VARCHAR(255)	NN	Regulatory rule title
description	TEXT	NN	Short purpose and scope summary
rule_text	TEXT	NN	Full statutory regulatory requirement
standard_disclosure	TEXT	OPT	Mandatory boilerplate text required
embedding_json	TEXT	OPT	Vector embedding (JSON float array) for RAG matching
created_at	DATETIME	NN	Rule insertion timestamp

10. precedent_submissions		Historical approved/rejected precedents for few-shot RAG context	
Column	Type	Modifiers	Description
id	VARCHAR(36)	PK NN	UUIDv4 precedent primary key
title	VARCHAR(255)	NN	Precedent document title
document_type	Enum(DocumentType)	NN (Default: other)	Classification type
masked_text	TEXT	NN	De-identified excerpt / document text
decision	Enum(DocumentStatus)	NN	Historic review verdict
officer_comment	TEXT	NN	Officer reasoning and historical notes
source_document_id	VARCHAR(36)	OPT	Original source document UUID (if converted)
embedding_json	TEXT	OPT	Vector embedding (JSON float array) for few-shot RAG
created_at	DATETIME	NN	Precedent record timestamp

6. Referential Integrity & Cascade Matrix

Parent Entity	Child Entity	Foreign Key Column	On Delete Behavior
users	documents	documents.advisor_id	Restricted (Preserves submission history)
users	review_decisions	review_decisions.officer_id	Restricted (Preserves audit verdict history)
users	notifications	notifications.user_id	Cascade Delete (all, delete-orphan)
documents	documents	documents.parent_document_id	Self-referencing tree (Set NULL / hierarchy)
documents	ai_analyses	ai_analyses.document_id	Cascade Delete (all, delete-orphan)
documents	review_decisions	review_decisions.document_id	Cascade Delete (all, delete-orphan)
documents	audit_events	audit_events.document_id	Cascade Delete (all, delete-orphan)
documents	pii_mappings	pii_mappings.document_id	Cascade Delete (all, delete-orphan)
ai_analyses	compliance_flags	compliance_flags.analysis_id	Cascade Delete (all, delete-orphan)